

# Bargaining and Timing Games

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# Infinite dynamic games

- Infinite dynamic games are hard to analyze without simplifying assumptions.
- Exception:
  - some bargaining models;
  - timing games with simple action space (e.g. stop/continue);

# Ultimatum Game: One Stage Bargaining

- One Proposer (P) and one Receiver (R);
- There is a surplus  $S = 1$  to be divided between the two;
- A failure to reach an agreement results in  $(\bar{u}_P, \bar{u}_R)$  with  $1 > \bar{u}_P + \bar{u}_R$ ;
- Timing: one period with sequential moves
  - 1 P offers a share  $\psi \in [0, 1]$  to R.
  - 2 R either:
    - accepts: resulting in  $(u_P, u_R) = (1 - \psi, \psi)$ .
    - rejects: resulting in  $(u_P, u_R) = (\bar{u}_P, \bar{u}_R)$ .
- There is no SPNE, in which  $\psi > \bar{u}_R$ .
- The SPNE involves  $(u_P^*, u_R^*) = (1 - \bar{u}_R, \bar{u}_R)$ .

# Finitely Repeated Rubinstein-Stahl Bargaining

- Consider the following extension of the Ultimatum Game:
  - there are  $T + 1$  (even) periods numbered  $\{0, 1, \dots, T\}$ ;
  - player 1 and 2 are negotiating over a division of a pie of size 1;
  - player 1 proposed a division  $(x_t, 1 - x_t)$  in even periods  $\{0, 2, \dots, T - 1\}$  where  $x_t$  is his share of the pie;
  - if player 2 accepts-game ends;
  - if player 2 rejects, then in next period he makes a counter offer  $(x_{t+1}, 1 - x_{t+1})$ ;
  - they alternate until they reach an agreement or period  $T + 1$ ;
  - if they fail to reach an agreement in period  $T + 1$ , each one gets 0.
  - player  $i$ 's discount rate is  $\delta_i$ .
- The SPNE is derived by backward induction.

# RS Bargaining Backward Induction

- Stage  $T$ , 2 proposes  $(0, 1)$  and 1 accepts;
- Stage  $T - 1$ , 1 proposes  $(1 - \delta_2, \delta_2)$  and 2 accepts;
- Stage  $T - 2$ , player 2 proposes  $(\delta_1(1 - \delta_2), 1 - \delta_1(1 - \delta_2))$  and 1 accepts;
- .....

## Proposition

Let  $T$  be an odd non-negative integer. The the Rubinstein-Stahl game has a unique SPNE, in which player 1 proposes his share to be

$$x_0^* = (1 - \delta_2) \frac{1 - (\delta_1 \delta_2)^{(T+1)/2}}{1 - \delta_1 \delta_2}$$

in period 0 and player 2 accepts the offer.

# Proof

Proof follows by induction:

- 1 For  $T = 1$ ,  $x_0^* = 1 - \delta_2$ . This corresponds to player 2 proposing  $(x_1^*, 1 - x_1^*) = (0, 1)$  in the last period. In stage 0, player 1 will make player 2 just indifferent between accepting and rejecting, making an offer of  $(x_1^*, 1 - x_1^*) = (1 - \delta_2, \delta_2)$ .
- 2 Assume that  $x_0^* = (1 - \delta_2) \frac{1 - (\delta_1 \delta_2)^{(K-1)/2}}{1 - \delta_1 \delta_2}$  is the solution of the game with  $T = K - 2$  ( $K - 1$  periods) where  $K$  is an odd integer.
- 3 We need to prove that  $x_0^*$  is the solution of the game with  $T = K$  ( $K + 1$  periods). Consider the subgame starting in period  $t = 2$ . It has  $K - 1$  periods and by our inductive assumption has a unique SPNE  $x_2^* = (1 - \delta_2) \frac{1 - (\delta_1 \delta_2)^{(K-1)/2}}{1 - \delta_1 \delta_2}$ .

## Proof Continued

By backward induction, at  $T = 1$ , the unique SPNE is for player 2 to propose a share  $x_1^* = \delta_1 x_2^*$  and for player 1 to accept. Folding the game to  $T = 0$ , player 1 will offer a share  $x_0^* = 1 - \delta_2(1 - \delta_1 x_2^*)$ . Plugging in for  $x_2^*$  we obtain:

$$\begin{aligned} x_0^* &= 1 - \delta_2 + \delta_1 \delta_2 (1 - \delta_2) \frac{1 - (\delta_1 \delta_2)^{(K-1)/2}}{1 - \delta_1 \delta_2} \\ &= (1 - \delta_2) \frac{1 - (\delta_1 \delta_2)^{(K+1)/2}}{1 - \delta_1 \delta_2} \end{aligned}$$

# Infinitely Repeated Rubinstein-Stahl Bargaining

- We cannot apply backward induction anymore.
- However, from player 1's perspective, the game beginning in stage 2 is identical to the game beginning in stage 0.
- In general, for player  $i$  the game beginning in stage  $k$  will be identical to the game beginning in stage  $k + 2$  where:
  - $k$  is even for player 1;
  - $k$  is odd for player 2;
- Let  $\bar{v}_i$  and  $\underline{v}_i$  be the highest and the lowest payoff respectively for player  $i$  in any SPNE;

# Infinitely Repeated Rubinstein-Stahl Bargaining

- Consider a subgame in which player  $i$  makes the offer.
- Player  $j$  will not accept an offer less than  $\delta_j \underline{v}_j$

$$\bar{v}_i \leq 1 - \delta_j \underline{v}_j \quad (1)$$

- Likewise, player  $j$  will accept any offer with at least  $\delta_j \bar{v}_j$

$$\underline{v}_i \geq 1 - \delta_j \bar{v}_j \quad (2)$$

- Subtracting (2) from (1)

$$\bar{v}_i - \underline{v}_i \leq \delta_j (\bar{v}_j - \underline{v}_j) \quad (3)$$

- Looking at a subgame with  $j$  making an offer results in

$$\bar{v}_j - \underline{v}_j \leq \delta_i (\bar{v}_i - \underline{v}_i) \quad (4)$$

# Infinitely Repeated Rubinstein-Stahl Bargaining

- Inequalities (3) and (4) result in

$$\bar{v}_i - \underline{v}_i \leq \delta_i \delta_j (\bar{v}_i - \underline{v}_i) \quad (5)$$

- This implies that  $\bar{v}_i = \underline{v}_i = v_i^* \Rightarrow$  the equilibrium is unique.
- Then, in the first round, it must be true that:

$$v_1 = 1 - \delta_2 v_2 \quad (6)$$

- If the game reaches the second round, it must be true that,

$$v_2 = 1 - \delta_1 v_1 \quad (7)$$

- Equations (6) and (7) uniquely pin down the equilibrium.

# Infinitely Repeated Rubinstein-Stahl Bargaining

## Proposition

The Rubinstein-Stahl Bargaining game with  $T = \infty$  has a unique SPNE which is described as follows. In any period in which  $i$  makes an offer he proposes his own share to be:

$$\frac{1 - \delta_j}{1 - \delta_i \delta_j}$$

and  $j$ 's share to be

$$\frac{\delta_j(1 - \delta_i)}{1 - \delta_i \delta_j}$$

- It turns out that the Rubinstein-Stahl Bargaining game is upper hemicontinuous (i.e.  $\lim_{T \rightarrow \infty} \Gamma^T(\delta_1, \delta_2) = \Gamma^\infty(\delta_1, \delta_2)$ ).

# Infinitely Repeated Rubinstein-Stahl Bargaining

- Besides upper hemicontinuity, R-S bargaining game has some intuitive predictions
  - The more patient the player is, the more surplus he gets.
  - There is a first mover advantage  $\delta_1 = \delta_2 = \delta$  results in

$$v_1^* = \frac{1}{1 + \delta}, v_2^* = \frac{\delta}{1 + \delta}$$

- The first mover advantage disappears as the real time between offers goes to zero.
- To see this, define the discount factor as  $\delta = e^{-r\tau}$  where
  - $r$  is a continuous discount factor capturing rate of time preference.
  - $\tau$  is the time between offers.
- Then,

$$v_1 = \frac{1 - e^{-r\tau}}{1 - e^{-2r\tau}}$$

- Using L'Hopital's Rule,  $\lim_{\tau \rightarrow 0} = \frac{1}{2}$ .

# War of Attrition

- War of Attrition is a timing game, in which players have two possible actions in each period (fight, back down).
- The game ends as soon as  $I - 1$  plays stop fighting.
- The last player standing gets the prize.
- First analyzed by Maynard Smith (Journal of Theoretical Biology, 1974).
- Economic application:
  - market application
  - *R&D* competition

# Discrete Time War of Attrition

- Two players ( $i = 1, 2$ );
- Infinite period  $t = 0, 1, 2, \dots$
- Fighting cost per period:  $c = 1$ ;
- Prize:  $v > 1$ ;
- Players choose simultaneously in each period: {fight, back down} =  $\{F, BD\}$ .
- Since it is costly to fight, boring NE always exists:
  - player  $i$  fight with probability one in each period;
  - player  $j$  backs down with probability one in period 0;
- Is this SPNE?

# Discrete Time War of Attrition

- We are interested in an equilibrium, in which players fight for a while;
- Players need to be mixing in such an equilibrium.
- Let  $p_i(t)$  be player  $i$ 's behavior strategy assigning probability  $p_i(t)$  to backing down in period  $t$ ;
- $p_i(t)$  solves the following Bellman equation:

$$\pi_i(t) = \max_{p_i(t)} [p_i(t)0 + (1 - p_i(t))(p_j(t)v + (1 - p_j(t))(\delta\pi_i(t+1) - 1))]$$

- In order for mixing to take place, the player needs to be indifferent between backing down and fighting.
- The expected payoff from not fighting is 0. Therefore,
  - $\pi_i(t) = \pi_i(t+1) = 0$
  - $p_j(t)v = (1 - p_j(t)) \implies p_j(t) = \frac{1}{1+v}$

# Discrete Time War of Attrition

- The equilibrium is stationary:  $p_i^*(t) = p_j^*(t) = \frac{1}{1+v}$ ;
- $Pr(F \text{ at periods } \{0, \dots, t-1\} \text{ and back down at } t) = (1-p)^t p = \left(\frac{v}{1+v}\right)^t \frac{1}{1+v}$ ;
- Note that player  $j$  has to mix with the right probability to make  $i$  indifferent between fighting and backing down;
- Simple extension:  $c_i \neq c_j$ ;
- Then  $p_i^* \neq p_j^*$ .
- Who will back down with higher probability in a mixed strategy equilibrium?

# Continuous Time War of Attrition

- $\delta = e^{-rt}$ ;
- cost of waiting for a length  $\epsilon$  is  $c(\epsilon) = \epsilon$ ;
- Player  $i$ 's strategy is described by the cdf  $G_i(t)$ .
- In order for player  $i$  to mix, in every instant he should be indifferent between fighting for another instant and backing down;
- At period  $t$ , player  $i$ 's benefit from fighting for another instant is:

$$\begin{aligned}\lim_{\epsilon \rightarrow 0} Pr_j(x \in (t, t + \epsilon) | x \geq t)v &= \lim_{\epsilon \rightarrow 0} \int_t^{t+\epsilon} \frac{dG_j}{1 - G_j(t)} \\ &= \lim_{\epsilon \rightarrow 0} \frac{G_j(t + \epsilon) - G_j(t)}{1 - G_j(t)} v\end{aligned}$$

# Continuous Time War of Attrition

- Equating the cost and benefit results in:

$$\lim_{\epsilon \rightarrow 0} \frac{G_j(\epsilon + t) - G_j(t)}{\epsilon} \frac{v}{1 - G(t)} = 1$$

$$\frac{g_j(t)}{1 - G_j(t)} = \frac{1}{v}$$

- Solving the differential equation results in:

$$G_j(t) = 1 - e^{-t/v}$$

- By symmetry,  $G_i(t) = G_j(t) = 1 - e^{-t/v}$ ;